

LIVING MYCELIUM BIOSENSORS · PRE-IGNITION FOREST-FIRE EARLY WARNING

Artificial Intelligence & Signal-Intelligence Plan

An LSTM Sensor-Fusion Approach to Pre-Ignition Wildfire Detection
from Mycelial Bioelectric Activity

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2026

Companion deck to the APA 7 written plan (Mycellium-Aegis_AI_Plan_APA7.md)

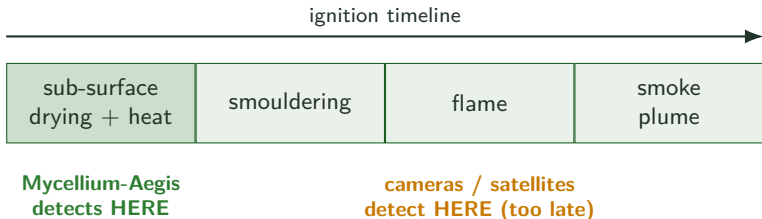
Where we are going

1. The Problem
2. Objectives & Success Criteria
3. Data Foundation
4. Method
5. Deployment & Governance
6. Risk & Roadmap
7. Conclusion

The Problem

Today's early-warning stack detects fire too late

Satellites, UAVs, camera towers and above-ground sensor networks all share one blind spot: they fire only **after** flame, heat radiation or smoke exists.



The shift

Move the detection window *earlier* — to the pre-ignition stage — by monitoring sub-surface moisture loss and heat build-up that precede ignition.

The core idea: listen to the forest floor

Underground **mycelial networks** produce reproducible, directional voltage **spikes** in response to heat, drying and damage.

At **15–20 cm** depth a $> 900^{\circ}\text{C}$ surface flame arrives as a damped $25\text{--}50^{\circ}\text{C}$ wave: cool enough for the mycelium to survive and keep signalling, yet coupled to the surface.

Adamatzky (2018, 2026a, 2026b); Phillips et al. (2023);
Doerr et al. (2025)

Depth	Effect on networks
0–2 cm	near-total destruction
2–5 cm	partial; spores survive
5–10 cm	pyrophilic fungi activate
15–20 cm	mycelium keeps sensing

The 15–20 cm “safe zone” is both survivable and measurable.

Why this is fundamentally an *AI* problem

The sensing is solved; the **decision** is the risk. Each raw channel is individually ambiguous:

- a hot afternoon raises soil temperature,
- a long drought raises substrate resistance,
- background metabolism produces occasional spikes.

Consequence

Any single channel, thresholded naively, produces false alarms that destroy operational trust and waste firefighting resources. The AI system exists to fuse weak, multi-rate signals into one **low-false-alarm** decision.

Objectives & Success Criteria

One headline commitment, measured

Primary objective

Detect fire onset in the pre-ignition phase with a **false-positive rate** $< 1\%$ while keeping recall high — because at hundreds–thousands of nodes, false alarms dominate operational trust.

Objective	Metric	Target
Minimize false alarms	FPR	$< 1\%$ (field-representative)
Preserve sensitivity	Recall	≥ 0.95
Balanced quality	PR-AUC / F_1	≥ 0.98 / ≥ 0.95
Timeliness	Latency	alarm before surface flame
Edge feasibility	Model / RAM	≤ 128 KB / ≤ 256 KB
Interpretability	Baseline agreement	explainable vs. AND rule

“Field-representative” = includes drought, diurnal/seasonal swings and non-fire heating.

Data Foundation

Experiment 1 — the signature is in the *frequency* domain

Single differential channel, *Pleurotus ostreatus*,
 $f_s \approx 45$ Hz, raw (GAIN_SIXTEEN,
 $7.81 \mu\text{V/bit}$).

	Resting	Fire stress
Mean voltage	-2.9 mV	-2.94 mV
Min voltage	-29.4 mV	-33.4 mV
Dominant freq.	≈ 20 Hz	≈ 5 Hz
Samples N	1,048	1,085

Key result

Mean voltage barely moves — but the
dominant band migrates $\approx 20 \rightarrow \approx 5$
Hz under heat stress.

A naive amplitude threshold fails; the
spectral and *distributional* change
carries the signal $\Rightarrow$ motivates
sequence / spectro-temporal
modelling.

Experiment 2 — toward field realism

Active soil–mycelium network, two channels, with real DSP.

Parameter	Experiment 1	Experiment 2
Subject	single tissue	soil/mycelium network
Channels	1 differential	2 differential (network map)
PGA gain	± 0.256 V	± 6.144 V ($24\times$ range)
Sample rate	≈ 45 Hz	≈ 100 Hz (Nyquist 50 Hz)
Filtering	none (raw)	5-tap linear-phase FIR [0.1, 0.2, 0.4, 0.2, 0.1]
Offset cal.	none	automatic (50-sample baseline)

FIR: unity DC gain ($\sum h_k = 1.0$), linear phase —
denoises without distorting the derivatives.

Two channels map thermal-stress *propagation*, not a
single point.

Labels, provenance & class imbalance

Four physical regimes (not just alarm/no-alarm):

- **normal** — diurnal/seasonal baseline
- **drought** — slow drying, no heat inflow
- **non-fire heating** — solar / sub-surface source
- **fire onset** — heat + rapid drying + spiking

This is what lets the model learn *drought vs. fire*.

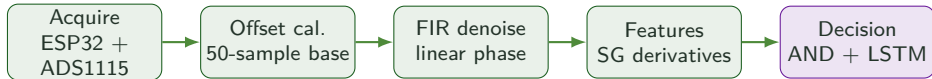
Imbalance (onset is rare) handled at three levels:

1. collect — over-sample controlled onsets;
2. resample — SMOTE on the *train split only*;
3. physics-informed augmentation (time-warp, jitter, baseline wander, noise superposition).

Provenance versioned per datasheet discipline (Gebru et al., 2021).

Method

Signal-processing & feature pipeline



Core fusion features

- heat-flux rate dT/dt
- drying rate $d(\ln R)/dt$
- spike frequency f_{spike}

Savitzky–Golay (1964) smooths signal *and* derivative in one pass.

Spectral & statistical

- band-power ratio, spectral centroid, dominant freq.
- variance, skew, kurtosis, Hjorth

A two-tier decision system

Tier 1 — deterministic (safety floor)

$\text{Alarm} = (dT/dt > \beta) \wedge (d(\ln R)/dt > \gamma) \wedge \text{spike}$

Interpretable, near-zero compute, always available. β, γ fixed at **98%** confidence in a climate chamber.

Tier 2 — learned (LSTM)

Learns the *joint temporal shape* the rigid AND rule ignores; outputs a calibrated probability with a **tunable operating point** for $\text{FPR} < 1\%$.

Escalation

High-severity alarm = LSTM probability over threshold **AND** deterministic rule corroborates **AND** neighbouring nodes agree. The interpretable rule stays in the loop; the model removes false alarms.

Model architecture — why recurrent

Fire onset is defined by how dT/dt , $d(\ln R)/dt$ and spike dynamics *co-evolve* over seconds–minutes, plus the spectral migration. LSTMs learn temporal dependencies while mitigating vanishing gradients (Hochreiter & Schmidhuber, 1997).

Reference model (edge-bounded)

- **Input:** windowed feature vector (fusion + spectral + statistical); optional raw channels.
- **Core:** 1–2 LSTM layers, 32–64 units; dropout regularizes the small-data regime.
- **Head:** 4-way softmax (normal / drought / non-fire heat / fire onset).
- **Calibration:** temperature scaling $\Rightarrow$ threshold controls FPR.

Sized against the ESP32 budget, not for maximal accuracy.

The LSTM is a baseline, not a foregone conclusion

Model family	Strength	Risk / fit
LSTM (baseline)	long-range temporal	sequential; primary choice
1D CNN	cheap, learns spectral motifs	weak long memory; strong front end
CNN-LSTM hybrid	motifs + evolution	larger; if either underfits
TCN	long receptive field, parallel	newer MCU export (Bai et al., 2018)
Transformer	global context	data-hungry; deferred (Vaswani et al., 2017)
Gradient-boosted trees	robust tabular baseline	no native temporal model

Selection rule

Whichever meets the targets at the **lowest edge cost**; interpretability breaks ties. All learned models are benchmarked against the deterministic rule *and* a gradient-boosted-trees baseline.

Training methodology

Honest splits

- grouped, time-blocked CV
- leave-one-session-out (lab)
- leave-one-site-out (field)
- no window straddles a split $\Rightarrow$ no leakage

Imbalance

- class-weighted / focal loss (Lin et al., 2017)
- SMOTE on train split only (Chawla et al., 2002)

Optimization

- Adam (Kingma & Ba, 2015)
- dropout (Srivastava et al., 2014), weight decay, grad clipping
- early stop on validation PR-AUC

The $< 1\%$ constraint

Enforced at *threshold-selection* time: pick the operating point on validation to hit $\text{FPR} < 1\%$ at max recall, freeze, then test once.

Evaluation & validation

Metrics (imbalance-aware)

- FPR (primary constraint), recall, precision, F_1
- PR-AUC (headline), ROC-AUC
- detection latency per burn event
- **reported per regime & per confounder**

Accuracy is meaningless under extreme imbalance.

Validation ladder

1. **Lab** (WP-2): climate chamber, radiant shock; fix β, γ .
2. **Field** (WP-4): buried nodes, controlled burn — alarm *before* flame.
3. **Spatial**: neighbour corroboration over LoRa suppresses idiosyncratic FPs.

Deployment & Governance

Edge deployment — decide on the node

Connectivity is intermittent and energy-expensive $\Rightarrow$ the hot-path decision runs **on the ESP32**; only confirmed events go over LoRa.

Compression

- INT8 post-training quantization (+ pruning if needed)
- TFLite-Micro runtime (Warden & Situnayake, 2019)
- release **gated** on post-quantization FPR/recall

Event-driven inference

- cheap always-on gate = deterministic checks + spike detect
- wake the LSTM only on a plausible anomaly
- average power stays within the 5 W solar + LiFePO₄ budget

MLOps, governance & responsible AI

Lifecycle

- data + features + code + model versioned together
- model cards (Mitchell et al., 2019); dataset datasheets
- drift monitoring → review → retrain → staged OTA
- Tier-1 rule = always-available fallback / rollback

Aligns with the BIGG Green Growth themes — climate, smart infrastructure, circular economy (TÜBİTAK, 2023).

Responsible AI

- false-positive discipline is the design axis
- transparency: evidence-carrying alarms
- human authority — the AI advises, never dispatches
- environmental integrity: edge compute + biodegradable, pyrophilic substrate

Risk & Roadmap

AI-specific risk register

Risk	Mitigation
False positives	3-signal AND; 4-class labels; threshold FPR < 1%; neighbour check
Missed onset	recall-preserving threshold; focal loss; deterministic floor
Small-data overfit	physics-informed augmentation; dropout; grouped CV; compact model
Lab→field shift	leave-one-site-out; drift monitoring; field retraining
Quantization loss	post-INT8 metric gate blocks release on regression
Sensor degradation	drift detection; Ag/AgCl or Pt-carbon electrodes; retrain
Automation over-reliance	evidence-carrying alarms; human authority retained

Complements — does not replace — the hardware/biological risk register.

Every deliverable maps to a funded work package

WP	Window	AI deliverable
WP-2	M4–M7	fix β, γ @98%; labelled onsets; first LSTM; per-regime eval
WP-3	M6–M9	FIR+SG denoise; event-gated inference; INT8 on ESP32
WP-4	M9–M11	field onsets; leave-one-site-out; latency-before-flame
WP-5	M11–M15	production LSTM @ FPR < 1%; model cards; TÜRKPATENT filing
WP-6	M15–M18	reference deployment; OTA updates & monitoring

Months relative to “Month 0” (Excellence Seal). WP-1 (M1–M3) builds the housings that generate the data.

Conclusion

From an imaging problem to an inference problem

- **Science established:** mycelium answers heat with a reproducible signature — decisively, the $\approx 20 \rightarrow \approx 5$ Hz spectral shift.
- **Engineering established:** at 15–20 cm, fire arrives as a survivable 25–50 °C wave — the signature is measurable.
- **The plan solves the decision problem:** an interpretable AND rule wrapped by an LSTM tuned to $\text{FPR} < 1\%$, on-device, governed, and validated per regime.

The commitment, designed in from line one

Earn operational trust by *not crying wolf* — a path from lab evidence (TRL 3→4) to a field-validated prototype (TRL 6→7) that is concrete and auditable.

Mycellium-Aegis — hearing the forest before it burns.

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Full APA-7 list (20 entries) in `Mycellium-Aegis_AI_Plan_APA7.md`.

Thank you

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